

a Source-class certificate ledger

Class L | Locally constant
proved IDs: L001
fixed finite memory becomes finitely many edge readouts in V

1/1

Class M | Multiplier
proved IDs: M001, M002, M003, M004, M005
 $q^2 = \lambda \bar{\lambda}$ is a fixed-support S -unit

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Class A | Algebraic action
proved IDs: A001, A002
regular algebraic action and endpoint shifts stay algebraic

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b Terminal upstream closure

	Frozen	Manifest	Pipeline	Record
Paper 3: multiplier	✓	✓	✓	✓
Paper 4: action	✓	✓	✓	✓

c Registered gates (9/9)

Escape semantics	✓	PASS	Exact controls	✓	PASS
Source lock	✓	PASS	Executable isolation	✓	PASS
Independent review	✓	PASS	Upstream bindings	✓	PASS
Proof ledger	✓	PASS	Output scope	✓	PASS
Scope ledger	✓	PASS			

Proof: 20 IDs / 0 cycles

Scope: 10 in / 9 out

Controls: 6 pass

Isolation: 12 files / 0 findings